RATIONAL SURFACES WITH SMOOTH CURVE GERMS BUT NO SMOOTH RATIONAL CURVES

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Abstract. We construct a normal projective rational surface whose only singular point is a rational singularity and whose anti-canonical divisor is ample, such that any closed point admits a smooth curve germ but the surface contains no smooth rational curve. Thus we answer negatively Question 3.5 of Zhuang's paper on smooth rational curves on singular rational surfaces.

Contents

1. Introduction 1
2. Preliminaries 4
3. The weighted hypersurface 6
4. A second example 11
References 15

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Zhuang [Zhu18] studies smooth rational curves on singular rational surfaces. A smooth projective rational surface is a blow-up of $\mathbb{P}^2$ or of a Hirzebruch surface [Har77, Corollary V.5.4, Theorem V.5.8, Example V.5.8.2, Remark V.5.8.4], and it contains many smooth rational curves. On a singular surface X there are two obstructions to the existence of a smooth rational curve. First, a singular point of X may admit no smooth curve germ (see Definition 2.1), as happens at an E_8 singularity. Second, the smooth locus of X may contain no rational curve at all. The second obstruction does not occur on a log del Pezzo surface of Picard number one, since the smooth locus of such a surface is dominated by rational curves [KM99, Theorem 1.3.1], which need not be smooth.

For surfaces with only quotient singularities, Zhuang answers the existence question positively: if X is a normal proper surface with only quotient singularities such that K_X is not pseudo-effective and any closed point of X admits a smooth curve germ, then X contains a smooth rational curve [Zhu18, Corollary 2.16]. Indeed, any normal proper surface with only quotient singularities such that K_X is not pseudo-effective and which contains no smooth rational curve has an E_8 singularity [Zhu18, Theorem 2.15 and the paragraph before it]. The complete local ring of an E_8 singularity is factorial [Lip69, Theorem 25.1], so an E_8 singularity admits no smooth curve germ, see also [GSLJ97, Introduction]. The E_8 singularity is the only quotient surface singularity that admits no smooth curve germ [GSLJ94], see also [Zhu18, Introduction].

Part of the argument in [Zhu18, Section 2] works for arbitrary rational singularities. On a rational surface X with only rational singularities, an integral curve $C \subset X$ is a smooth rational curve if and only if $|K_X + C| = \emptyset$ [Zhu18, Corollary 2.3]. Moreover, if X is a normal proper surface with only rational singularities and K_X is not pseudo-effective, then X contains no

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smooth rational curve if and only if $-K_X$ is effective and $\mathrm{Cl}(X)$ is infinite cyclic, generated by the class of $-K_X$ [Zhu18, Proposition 2.5].

The assumption that the singularities are quotient singularities is used only after [Zhu18, Proposition 2.5], and it enters through the local class groups. The local class group at a quotient singularity is the abelianization of the finite subgroup of $\mathrm{GL}(2,\mathbb{C})$ defining it [Zhu18, Lemma 2.7]. For a surface with only quotient singularities satisfying the conditions of [Zhu18, Proposition 2.5], each local class group is cyclic of order equal to the local Cartier index of K_X [Zhu18, Corollary 2.8], and E_8 is the only Du Val singularity in the list of possible singular points following [Zhu18, Corollary 2.8]. A rational singularity that is not a quotient singularity is not defined by a finite group, and no such list exists for it. On the local side, the rational surface singularities that contain a smooth curve were characterized in terms of the minimal resolution [GSLJ94, GSLJ97].

Zhuang also studies surfaces with a rational singularity that is not a quotient singularity. For any integer $m \geq 5$, he constructs in [Zhu18, Example 3.4] a normal proper surface X with rational singularities such that K_X is not pseudo-effective and X contains no smooth rational curve. The surface X has two singular points, an E_8 singularity and a rational singularity that is not a quotient singularity. Because of the E_8 singularity, X does not satisfy the assumption on smooth curve germs. The surfaces constructed before [Zhu18, Theorem 2.15] and the surfaces in [Zhu18, Example 3.4] all start from a Gorenstein del Pezzo surface of Picard number one with an E_8 singularity. Zhuang blows up the double point of an anti-canonical rational curve contained in the smooth locus and points infinitely near it, and then contracts the strict transforms of the anti-canonical curve and of all exceptional curves except the last one to one point. The E_8 singularity is not affected. Zhuang remarks after [Zhu18, Example 3.4] that the surfaces of [Zhu18, Example 3.4] do not seem to have many variants, and asks the following question.

Question 1.1 ([Zhu18, Question 3.5])**.** Let X be a normal proper surface with rational singularities. Assume that K_X is not pseudo-effective and that any closed point of X admits a smooth curve germ (see Definition 2.1). Does X contain a smooth rational curve, that is, a closed curve isomorphic to $\mathbb{P}^1$?

In [Zhu18, Question 3.5], Zhuang also asks for a classification of all surfaces with rational singularities that contain no smooth rational curve.

Let X be a surface satisfying the assumptions of Question 1.1 and containing no smooth rational curve. Then X has a singular point that is rational but not a quotient singularity by [Zhu18, Corollary 2.16]. In particular, X does not satisfy the assumption of quotient singularities in [KM99, Theorem 1.3.1]. Moreover, $-K_X$ is effective and $\mathrm{Cl}(X)$ is infinite cyclic, generated by the class of $-K_X$, by [Zhu18, Proposition 2.5], and $-K_X$ is ample and X has Picard number one by [Zhu18, Lemma 2.4 and the paragraph after it]. Since a normal surface singularity is log terminal if and only if it is a quotient singularity [KM98, Proposition 4.18], the singular point of X that is not a quotient singularity is unique [PP24, Proposition B]. The goal of this paper is to answer Question 1.1 negatively. In Theorem 1.2 we construct a normal projective rational surface X with a unique singular point such that X satisfies the assumptions of Question 1.1 and contains no closed curve isomorphic to $\mathbb{P}^1$.

We prove the following theorem.

Theorem 1.2. *Let* $\mathbb{P}(1,4,3,5) = \operatorname{Proj}\mathbb{C}[u,x,z,t]$ *be the weighted projective space, where* $\deg u = 1$, $\deg x = 4$, $\deg z = 3$, *and* $\deg t = 5$. *Let* $X \subset \mathbb{P}(1,4,3,5)$ *be the hypersurface defined by*

$$(z^2 - ut)^2 = x^3 + u^9 z,$$

let $p := [0:0:0:1] \in X$, *and let* $D \subset X$ *be the reduced closed subscheme defined by* $u = 0$. *Then the following hold.*

(1) X is a normal projective rational surface, p is the unique singular point of X, and $X \ni p$ is a rational singularity.

(2) *D is a prime divisor, D is $\mathbb{Q}$-Cartier and ample,* $\mathrm{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$, *and* $-K_X \sim D$. *In particular, K_X is not pseudo-effective.*
(3) *Any closed point of X admits a smooth curve germ (see Definition 2.1).*
(4) *X contains no closed curve isomorphic to $\mathbb{P}^1$.*

Since X is projective, X is a normal proper surface with rational singularities. Thus Theorem 1.2 answers Question 1.1 negatively. An explicit smooth curve germ of X at p is given in Proposition 3.6.

In Section 4 we construct a second normal projective rational surface whose unique singular point is a rational singularity, such that any closed point admits a smooth curve germ, the class group is infinite cyclic and generated by an ample prime divisor in the anti-canonical class, and no closed curve is isomorphic to $\mathbb{P}^1$. It is obtained by blowing up $\mathbb{P}^2$ ten times and contracting ten smooth rational curves to one point (Construction 4.1, Construction-Proposition 4.3, and Proposition 4.4).

Sketch of the proof. By Lemma 2.4, a consequence of [Zhu18, Proposition 2.5], it suffices to prove (1) and (3) of Theorem 1.2 and that D is a prime divisor with $\mathrm{Cl}(X) = \mathbb{Z}[D]$ and $K_X \sim -D$. By Lemma 3.2, D is a prime divisor, and $X \backslash D \cong \mathbb{A}^2$ with coordinates $a = x/u^4$ and $b = (z^2 - ut)/u^6$, so $\mathrm{Cl}(X) = \mathbb{Z}[D]$, and $K_X \sim -D$ because the two-form $da \wedge db$ has a pole of order one along D (Lemma 3.4). The point p is the unique singular point of X (Lemma 3.3), and $X \ni p$ is a rational singularity since X is rational and $H^0(X, \mathcal{O}_X(K_X)) = H^0(X, \mathcal{O}_X(-D)) = 0$ (Lemma 3.5). For (3) at p (Proposition 3.6), the chart $(t \neq 0)$ is V/μ_5, where $V \subset \mathbb{A}^3$ is the hypersurface

$$(z^2 - u)^2 = x^3 + u^9 z,$$

μ_5 acts with weights $(1, 4, 3)$, and p is the image of 0 (Lemma 3.3). The curve $V \cap (z = 0)$ is the cusp $u^2 = x^3$ with normalization $s \to (u, x) = (s^3, s^2)$, on which $\zeta \in \mu_5$ acts by $s \to \zeta^2 s$. Since $s^5 = ux$ is a function on the cusp, the coordinate ring of the image of the cusp in V/μ_5 is

$$\mathbb{C}[s^2, s^3]^{\mu_5} = \mathbb{C}[s]^{\mu_5} = \mathbb{C}[s^5],$$

so the image is smooth at p.

The surface X and the germ at p are explicit, while (4) is proved without determining the rational curves on X. Apart from Lemma 2.4, the proof consists of computations on the charts $(u \neq 0)$, $(x \neq 0)$, and $(t \neq 0)$ and one Leray argument. A resolution of p is not computed. The new ingredients are the polynomial F of Construction 3.1 and the smooth curve germ at p in Proposition 3.6.

(a) The chart of weight 5. A surface answering Question 1.1 negatively has a singular point that is rational but not a quotient singularity [Zhu18, Corollary 2.16], and K_X is not Cartier near it, since a rational surface singularity near which K_X is Cartier is Du Val, hence a quotient singularity [KM98, Theorem 4.20, Corollary 5.24, Remark 4.21(1)]. We take X to be the closure in $\mathbb{P}(1, 4, 3, 5)$ of the image of $\mathbb{A}^2$ under

$$(a, b) \to [1 : a : b^2 - a^3 : (b^2 - a^3)^2 - b]$$

(Lemma 3.2), so that $X \backslash D$ is the affine plane in the chart $(u \neq 0)$. The singular point p is not in the chart $(u \neq 0)$, on which X is a hypersurface in $\mathbb{A}^3$ and K_X is Cartier, but in the chart $(t \neq 0)$ of weight 5, where K_X is not Cartier near p (Lemma 3.3).

(b) The surface of Section 4. After the first eight blow-ups of Construction 4.1, the curves $L, E_1, \ldots, E_7$ form a configuration of (-2)-curves of type E_8, whose contraction is a Gorenstein del Pezzo surface of Picard number one with an E_8 singularity, and the only curve with coefficient one in the anti-canonical divisor $3L + 2E_1 + 4E_2 + 6E_3 + 5E_4 + 4E_5 + 3E_6 + 2E_7 + E_8$ is E_8, outside the configuration. (Neither the coefficients nor the contraction are used in any proof, so we only indicate their sources. The coefficients follow from the blow-up formula, by the computation in the proof of Lemma 4.2(3) stopped after the eighth blow-up. The contraction exists by [Lip69, Theorem 27.1], since the proofs of Construction-Proposition 4.3, except the step

showing $\mathrm{mult}_{E_8} F_p = 1$, and of Proposition 4.4(1)(2) apply to the surface after the first eight blow-ups, with $L + E_1 + \cdots + E_7$, E_8, and $3L + 2E_1 + 4E_2 + 6E_3 + 5E_4 + 4E_5 + 3E_6 + 2E_7$ in place of T, E_{10}, and G. Its singular point is Du Val [KM98, Definition 4.4, Theorem 4.20], since the contracted curves are (-2)-curves, and its anti-canonical divisor is ample, since its Picard number is one and $((3L + 2E_1 + \cdots + E_8) \cdot E_8) = 1$.)

The constructions in [Zhu18, Theorem 2.15 and the paragraph before it, Example 3.4] also start from such a surface, but they blow up points over its smooth locus, so the E_8 singularity remains. The last two blow-ups are over the E_8 singularity. They move E_8 into the divisor $T = L + E_1 + \cdots + E_9$, which is contracted, and the blow-up Y satisfies $-K_Y \sim E_{10} + G$, where $G \geq 0$ is the divisor of Construction 4.1 supported on T, with $\mathrm{mult}_{E_8} G = 1$ (Lemma 4.2(3)). Since $(G \cdot C) \leq 0$ for any irreducible component C of T, the fundamental cycle of T (Definition 2.2) is at most G. Since $K_Y + G \sim -E_{10}$, the upper bound G for the fundamental cycle of T implies the positivity of the Euler characteristic in [Lip69, Theorem 27.1], so T contracts to a rational singularity p of a normal projective surface X, and it implies $\mathrm{mult}_{E_8} F_p = 1$ for the fundamental cycle F_p of $X \ni p$ on Y, which is the fundamental cycle of T (Construction-Proposition 4.3). Thus p admits a smooth curve germ by Lemma 2.3, and Lemma 2.4 excludes closed curves isomorphic to $\mathbb{P}^1$ (Proposition 4.4).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce smooth curve germs and the fundamental cycle, and deduce from a criterion of Zhuang [Zhu18, Proposition 2.5] that a projective surface with rational singularities whose class group is generated by an anti-canonical prime divisor contains no smooth rational curve (such a surface is normal [KM98, Definition 5.8], and the class of a prime divisor on a normal projective surface is not torsion, so the class group is infinite cyclic, and the claim follows from Lemma 2.4). In Section 3 we prove Theorem 1.2. In Section 4 we construct the second example and prove Proposition 4.4.

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2. Preliminaries

In this section we collect two tools. Lemma 2.3 detects a smooth curve germ at a rational surface singularity from one coefficient of the fundamental cycle, and it is used in Section 4. Lemma 2.4 excludes closed curves isomorphic to $\mathbb{P}^1$ by the class group, and it is used in Sections 3 and 4.

2.1. Smooth curve germs and the fundamental cycle.

Definition 2.1. Let X be a normal surface and let $p \in X$ be a closed point. A *smooth curve germ* of X at p is a reduced closed analytic subspace germ $(\Gamma, p) \subset (X, p)$ of dimension one such that the local analytic ring $\mathcal{O}_{\Gamma,p}$ is isomorphic to the ring $\mathbb{C}\{s\}$ of convergent power series in one variable. We say that p *admits a smooth curve germ* if there exists a smooth curve germ of X at p.

Any smooth point p of X admits a smooth curve germ: if s_1, s_2 are local analytic coordinates of X at p, then the germ defined by s_2 is a smooth curve germ of X at p. A smooth curve germ of X at p need not be defined by one equation, and it need not be the germ of a closed curve on X. In fact, at a singular point p of X, no smooth curve germ is defined by one equation: if the ideal of a smooth curve germ (Γ, p) in the local analytic ring of X at p, or in the completion of that ring, is generated by one element f, then the maximal ideal of the ring is generated by f and a lift of a coordinate of Γ at p, so the ring is regular and p is a smooth point of X. Thus X admits no smooth curve germ at a singular point p whose complete local ring is a UFD. Indeed,

the ideal of a smooth curve germ in the complete local ring of X at p is a prime ideal of height one, because the quotient ring is the ring of formal power series in one variable and the complete local ring is a two-dimensional complete local domain, hence catenary [Mat89, Theorem 29.4(ii), Theorem 31.4]. Thus the ideal is principal, since a prime ideal of height one in a UFD is principal [Mat89, Theorem 20.1]. The E_8 singularity is such a point [Lip69, Theorem 25.1].

The term smooth curve germ is used in [Zhu18] without a definition. The condition in [Zhu18, Corollary 2.16] is that for any closed point p there exists a smooth curve (locally analytically) passing through p, and Definition 2.1 is our reading of the words "a smooth curve (locally analytically) passing through p".

Definition 2.2. Let Y be a smooth surface and let $E_1, \dots, E_k \subset Y$ be distinct proper integral curves such that $E_1 \cup \cdots \cup E_k$ is connected and the intersection matrix $((E_i \cdot E_j))_{1 \le i,j \le k}$ is negative definite. The *fundamental cycle* of $E_1 + \cdots + E_k$ is the smallest non-zero divisor $F = \sum_{i=1}^k a_i E_i$ with integers $a_i \ge 0$ such that $(F \cdot E_i) \le 0$ for any i, that is, $F \le F'$ for any non-zero divisor $F' = \sum_{i=1}^k b_i E_i$ with integers $b_i \ge 0$ such that $(F' \cdot E_i) \le 0$ for any i. Such F exists, and $a_i \ge 1$ for any i [Art66, Proposition 2 and pp. 131–132]. If $\pi : Y \to X$ is a resolution of singularities of a normal surface X which is an isomorphism over the smooth locus of X, and $p \in X$ is a singular point, then $\pi^{-1}(p)$ is connected [Lip69, p. 275], it is a curve since it is not finite by Zariski's Main Theorem and p is singular (cf. [Har77, Theorem V.5.2]), and the intersection matrix of the irreducible components of $\pi^{-1}(p)$ is negative definite [KM98, Lemma 3.40]. In this case, we call the fundamental cycle of the sum of the irreducible components of $\pi^{-1}(p)$ the *fundamental cycle* of $X \ni p$ on Y, and denote it by F_p.

Definition 2.2 is written for any connected union of proper integral curves on a smooth surface whose intersection matrix is negative definite, and not only for the curves over a singular point, because in Section 4 the fundamental cycle of T is used to verify the conditions for the contraction in Construction-Proposition 4.3, before the contraction exists.

The following lemma reduces the existence of a smooth curve germ at a rational singular point p to the condition that one coefficient of F_p is equal to 1.

Lemma 2.3. *Let X be a normal surface, let $\pi : Y \to X$ be a resolution of singularities which is an isomorphism over the smooth locus of X, and let $p \in X$ be a singular point such that $X \ni p$ is a rational singularity. Let F_p be the fundamental cycle of $X \ni p$ on Y. Assume that $\operatorname{mult}_E F_p = 1$ for some irreducible component E of $\pi^{-1}(p)$. Then p admits a smooth curve germ.*

Proof. Let $\mathfrak{m}_p$ be the maximal ideal of $\mathcal{O}_{X,p}$ and let $f_1, \dots, f_N$ be generators of $\mathfrak{m}_p$. By [Art66, Theorem 4] (see also [GSLJ97, 1.14] and [Lip69, Proposition 3.1, Theorem 12.1]),

$$\mathfrak{m}_p \mathcal{O}_Y = \mathcal{O}_Y(-F_p)$$

over a neighborhood of p. Let $q \in E$ be a smooth point of E which is not contained in any other irreducible component of $\pi^{-1}(p)$. Such points form a non-empty open subset of E. Since $\operatorname{mult}_E F_p = 1$, a local equation of F_p at q is a regular parameter w of $\mathcal{O}_{Y,q}$. Since $f_1 \circ \pi, \dots, f_N \circ \pi$ generate the ideal (w) of the local ring $\mathcal{O}_{Y,q}$, there exists j such that $f_j \circ \pi$ is w times a unit of $\mathcal{O}_{Y,q}$. Let (w, v) be a regular system of parameters of $\mathcal{O}_{Y,q}$ and let Γ' be the germ at q of the analytic curve $(v = 0)$. Then Γ' is a smooth analytic curve germ transverse to E at q, the restriction $w|_{\Gamma'}$ is a coordinate of Γ' at q, and hence $s := (f_j \circ \pi)|_{\Gamma'}$ is also a coordinate of Γ' at q.

The elements $f_1, \dots, f_N$ also generate the maximal ideal of the local analytic ring of X at p, as one sees on an affine neighborhood of p. Since a holomorphic map germ $(X, p) \to (\mathbb{C}^N, 0)$ whose components generate the maximal ideal of the local analytic ring of X at p is a closed embedding, the map $(f_1, \dots, f_N)$ embeds the germ (X, p) into $(\mathbb{C}^N, 0)$. Let $x_1, \dots, x_N$ be the coordinates of $\mathbb{C}^N$. Then $(f_i \circ \pi)|_{\Gamma'} = \varphi_i(s)$ for convergent power series $\varphi_i \in \mathbb{C}\{s\}$ with $\varphi_j(s) = s$, so $\pi(\Gamma')$ is the germ $\Gamma \subset (\mathbb{C}^N, 0)$ defined by $x_i = \varphi_i(x_j)$ for any $i \ne j$. Thus Γ is a reduced closed analytic subspace germ of (X, p) of dimension one with $\mathcal{O}_{\Gamma,p} \cong \mathbb{C}\{s\}$, that is, a smooth curve germ of X at p. □

The criterion of Lemma 2.3, together with the converse, is due to [GSLJ94] and [GSLJ97, Proposition 1.1, Corollary 1.2, 1.14], where it is formulated for formal curves on the spectrum of a complete local ring. We need analytic germs, so we include the short proof above.

2.2. **Surfaces without smooth rational curves.** The constructions of Sections 3 and 4 are designed to satisfy the hypotheses of the following lemma, which is the form in which we use the implication (2)⇒(1) of [Zhu18, Proposition 2.5].

Lemma 2.4. *Let X be a normal projective surface with rational singularities, and let D be a prime divisor on X. Assume that $-K_X \sim D$ and $\mathrm{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$. Then D is $\mathbb{Q}$-Cartier and ample, K_X is not pseudo-effective, and X contains no closed curve isomorphic to $\mathbb{P}^1$.*

Proof. Let A be an ample Cartier divisor on X. Since $\mathrm{Cl}(X) = \mathbb{Z}[D]$, there exists an integer n such that $A \sim nD$. Since A is ample, $(A \cdot D) > 0$, and

$$(A^2) = n(A \cdot D) > 0,$$

so $n > 0$. Thus $nD \sim A$ is Cartier and ample, and hence D is $\mathbb{Q}$-Cartier and ample.

Since $K_X \sim -D$, K_X is $\mathbb{Q}$-Cartier and $(K_X \cdot A) = -(D \cdot A) < 0$. A pseudo-effective class has non-negative intersection number with A, since the classes of divisors ≥ 0 do and the set of classes with non-negative intersection number with A is closed [Laz04, Definition 2.2.25]. Thus K_X is not pseudo-effective.

Finally, X has only rational singularities, K_X is not pseudo-effective, and $\mathrm{Cl}(X)$ is infinite cyclic and generated by the class of the divisor $D \geq 0$ with $D \sim -K_X$. By the implication (2)⇒(1) of [Zhu18, Proposition 2.5], whose hypotheses are that the singularities are rational and that K_X is not pseudo-effective, X contains no closed curve isomorphic to $\mathbb{P}^1$. □

3. The weighted hypersurface

In this section, we prove Theorem 1.2. The surface X of Theorem 1.2 is a compactification of the affine plane by one irreducible curve D, so the class group of X is generated by D. The main point of the section is the smooth curve germ at the unique singular point p of X, which is a point of D (Proposition 3.6).

The polynomial F of Construction 3.1 below is designed for two charts of $\mathbb{P}(1,4,3,5)$: in the chart $(u \neq 0)$ of weight 1, the open subset $X \backslash D$ is the affine plane with the coordinates a, b of Construction 3.1(5) (Lemma 3.2), and the chart $(t \neq 0)$ of weight 5 contains the unique singular point p (Lemma 3.3). On $X \backslash D$, the functions x/u^4, z/u^3, and t/u^5 are a, $b^2 - a^3$, and $(b^2 - a^3)^2 - b$, and the weights 4, 3, and 5 of x, z, and t are the orders of the poles of a, $b^2 - a^3$, and $(b^2 - a^3)^2 - b$ along D, since u vanishes to order one along D while x, z, t, and $z^2 - ut$ do not vanish identically on D. The degree 12 of F is the order of the poles of b^2 and of a^3 along D, and on $X \backslash D$ the term $u^9 z$ of F is $u^{12}(b^2 - a^3)$, so F expresses that the difference $b^2 - a^3$ has a pole of order 3 only.

Construction 3.1. Let $\mathbb{P}(1,4,3,5) = \operatorname{Proj} \mathbb{C}[u,x,z,t]$ with $\deg u = 1$, $\deg x = 4$, $\deg z = 3$, and $\deg t = 5$. Let F, X, p, D, a, b, V, and W be as follows.

(1) $F := (z^2 - ut)^2 - x^3 - u^9 z$, a weighted homogeneous polynomial of degree 12.
(2) $X := \operatorname{Proj} \mathbb{C}[u,x,z,t]/(F) \subset \mathbb{P}(1,4,3,5)$.
(3) $p := [0:0:0:1] \in X$.
(4) $D \subset X$ is the reduced closed subscheme defined by $u = 0$.
(5) $a := x/u^4$ and $b := (z^2 - ut)/u^6$, regular functions on $X \backslash D$.
(6) $V \subset \mathbb{A}^3$ is the hypersurface defined by $(z^2 - u)^2 - x^3 - u^9 z = 0$, where u, x, z are the coordinates of $\mathbb{A}^3$, and the group $\mu_5 \subset \mathbb{C}^*$ of fifth roots of unity acts on V by $\zeta \cdot (u,x,z) = (\zeta u, \zeta^4 x, \zeta^3 z)$ for $\zeta \in \mu_5$.
(7) $W \subset \mathbb{A}^3$ is the hypersurface defined by $(z^2 - ut)^2 - 1 - u^9 z = 0$, where u, z, t are the coordinates of $\mathbb{A}^3$, and the group $\mu_4 \subset \mathbb{C}^*$ of fourth roots of unity acts on W by $\zeta \cdot (u,z,t) = (\zeta u, \zeta^3 z, \zeta t)$ for $\zeta \in \mu_4$.

Lemma 3.2. *Notation as in Construction 3.1. Then F is irreducible, X is a normal projective surface, D is a prime divisor on X, and the morphism $(a,b) : X\backslash D \to \mathbb{A}^2$ is an isomorphism.*

Proof. Since $\deg u = 1$, the chart $(u \neq 0)$ of $\mathbb{P}(1,4,3,5)$ is the affine space with coordinates $a = x/u^4$, z/u^3, and t/u^5 [Ian00, 5.3]. Dividing F by u^{12}, we see that $X\backslash D = X \cap (u \neq 0)$ is defined by

$$b^2 - a^3 - \frac{z}{u^3} = 0, \qquad \text{where } b = \left(\frac{z}{u^3}\right)^2 - \frac{t}{u^5},$$

and we have

$$\mathbb{C}[a, z/u^3, t/u^5] = \mathbb{C}[a, z/u^3, b].$$

Thus the coordinate ring of $X\backslash D$ is $\mathbb{C}[a,b]$, and $(a,b) : X\backslash D \to \mathbb{A}^2$ is an isomorphism, with inverse given by $a = x/u^4$ and

$$\frac{z}{u^3} = b^2 - a^3, \qquad \frac{t}{u^5} = (b^2 - a^3)^2 - b.$$

We show that F is irreducible. The localization $\mathbb{C}[u,x,z,t]_u$ is the polynomial ring $\mathbb{C}[u,u^{-1}][a, z/u^3, t/u^5]$, and $F = u^{12}(b^2 - a^3 - z/u^3)$. Thus the computation above implies

$$\mathbb{C}[u,x,z,t]_u/(F) \cong \mathbb{C}[u,u^{-1},a,b],$$

which is a domain, so F is a prime element of $\mathbb{C}[u,x,z,t]_u$. Let $F = F_1F_2$ be a factorization in $\mathbb{C}[u,x,z,t]$. Then one of the factors, say F_1, is a unit in $\mathbb{C}[u,x,z,t]_u$, so $F_1 = cu^k$ for some $c \in \mathbb{C}^*$ and some integer $k \geq 0$. Since $F(0,x,z,t) = z^4 - x^3 \neq 0$, u does not divide F, so $k = 0$. Therefore, F is irreducible. In particular, $\mathbb{C}[u,x,z,t]/(F)$ is a domain, and X is integral.

We write F_u, F_x, F_z, and F_t for the partial derivatives of F. The affine cone $(F = 0) \subset \mathbb{A}^4$ is smooth on $(u \neq 0)$, since the coordinate ring of the cone localized at u is $\mathbb{C}[u,u^{-1},a,b]$. On $(u = 0)$, the equation $F = 0$ becomes $z^4 = x^3$, and $F_x = -3x^2$. Thus a singular point of the cone with $u = 0$ satisfies $x = 0$, and hence $z = 0$. Conversely, on the line $u = x = z = 0$ we have

$$F_u = -2t(z^2 - ut) - 9u^8z = 0, \qquad F_x = -3x^2 = 0,$$
$$F_z = 4z(z^2 - ut) - u^9 = 0, \qquad F_t = -2u(z^2 - ut) = 0.$$

Thus, by the Jacobian criterion [Har77, Theorem I.5.1], the singular locus of the cone is the line $u = x = z = 0$, which is of codimension two in the three-dimensional cone. Therefore, the cone is normal [Har77, Proposition II.8.23(b)], that is, $\mathbb{C}[u,x,z,t]/(F)$ is a normal graded domain. The surface X is covered by the open subsets $(u \neq 0)$, $(x \neq 0)$, $(z \neq 0)$, and $(t \neq 0)$, and the coordinate ring of each of them is the degree-zero part of the localization of $\mathbb{C}[u,x,z,t]/(F)$ at the corresponding coordinate. A localization of a normal graded domain is a normal graded domain, and the degree-zero part of a normal graded domain is normal, as an element of the fraction field of the degree-zero part that is integral over it belongs to the graded domain and is homogeneous of degree zero. Thus X is normal. Since $\mathbb{P}(1,4,3,5)$ is a projective variety [Ian00, 5.2] and X is a closed subscheme of $\mathbb{P}(1,4,3,5)$, X is projective. Since $X\backslash D \cong \mathbb{A}^2$ is a non-empty open subset of the integral scheme X, $\dim X = 2$.

Finally, the closed subscheme $(u = 0)$ of X is $\operatorname{Proj}\mathbb{C}[x,z,t]/(z^4 - x^3)$. Since $\gcd(3,4) = 1$, $z^4 - x^3$ is irreducible, so $\mathbb{C}[x,z,t]/(z^4 - x^3)$ is a domain of dimension two. Thus D is an integral curve on X, that is, D is a prime divisor on X. □

Lemma 3.3. *Notation as in Construction 3.1. Then μ_5 acts freely on $V\backslash\{0\}$, V is normal with $\operatorname{Sing} V = \{0\}$, and the open subset $X\backslash(t = 0)$ is isomorphic to V/μ_5, where p corresponds to the image of 0. Moreover, $\operatorname{Sing} X = \{p\}$, and 5 is the smallest positive integer m such that mK_X is Cartier near p.*

Proof. By [Ian00, 5.3], the chart $(t \neq 0)$ of $\mathbb{P}(1,4,3,5)$ is $\mathbb{A}^3/\mu_5$, where $\mathbb{A}^3$ has coordinates u, x, z and μ_5 acts as in Construction 3.1(6): the chart is obtained by setting $t = 1$, and μ_5 acts by

changing the choice of a fifth root of t. The intersection of X with the chart $(t \neq 0)$ is defined by $F(u,x,z,1) = (z^2-u)^2 - x^3 - u^9 z$, which is semi-invariant of weight 2:

$$F(\zeta u, \zeta^4 x, \zeta^3 z, 1) = \zeta^2 F(u,x,z,1) \quad \text{for any } \zeta \in \mu_5.$$

Thus V is μ_5-stable. Since taking invariants of the finite group μ_5 is exact, $X\backslash(t=0) \cong V/\mu_5$, and $p = [0:0:0:1]$ corresponds to the image of 0.

Let $\zeta \in \mu_5\backslash\{1\}$. Since 4 and 3 are coprime to 5, we have $\zeta^4 \neq 1$ and $\zeta^3 \neq 1$, so the only fixed point of ζ in $\mathbb{A}^3$ is 0. Thus μ_5 acts freely on $V\backslash\{0\}$, and the quotient morphism $V\backslash\{0\} \to (V\backslash\{0\})/\mu_5$ is étale.

We show that $\operatorname{Sing} V = \{0\}$. The partial derivatives of $F(u,x,z,1)$ with respect to u,x,z are F_u, F_x, F_z evaluated at $t=1$. Let (u,x,z) be a singular point of V. If $u=0$, then $F_x = -3x^2 = 0$ implies $x=0$, and the equation of V implies $z^4 = 0$, so $(u,x,z) = 0$. Suppose that $u \neq 0$. The weighted Euler identity

$$12F = uF_u + 4xF_x + 3zF_z + 5tF_t$$

at $(u,x,z,1)$ implies $F_t(u,x,z,1) = -2u(z^2-u) = 0$, so $z^2 = u$. Then

$$F_z(u,x,z,1) = 4z(z^2-u) - u^9 = -u^9 \neq 0,$$

a contradiction. Thus $\operatorname{Sing} V \subset \{0\}$. Since

$$F(u,x,z,1) = u^2 - 2uz^2 + z^4 - x^3 - u^9 z$$

has no linear term, 0 is a singular point of V, so $\operatorname{Sing} V = \{0\}$. Since V is a hypersurface whose singular locus is of codimension two, V is normal [Har77, Proposition II.8.23(b)].

We show that X is smooth outside p. By Lemma 3.2, $X\backslash D \cong \mathbb{A}^2$ is smooth. Since $z^4 = x^3$ on D, the coordinates x and z are both non-zero at any point $q \in D\backslash\{p\}$. Let $q \in D\backslash\{p\}$. By [Ian00, 5.3], the chart $(x \neq 0)$ of $\mathbb{P}(1,4,3,5)$ is $\mathbb{A}^3/\mu_4$, where $\mathbb{A}^3$ has coordinates u,z,t and μ_4 acts as in Construction 3.1(7), and $X \cap (x \neq 0)$ is W/μ_4, since $F(u,1,z,t)$ is μ_4-invariant. Let $w \in W$ be a point over q. Then $u=0$ and $z \neq 0$ at w. Since $\gcd(4,3) = 1$, $\zeta^3 = 1$ implies $\zeta = 1$ for $\zeta \in \mu_4$, so μ_4 acts freely on $(z \neq 0)$, and the quotient morphism $W \to X \cap (x \neq 0)$ is étale at w. At w, the Euler identity with $u=0$, $x=1$, and $F=0$ reads $4F_x + 3zF_z + 5tF_t = 0$. Since $F_x = -3x^2 = -3$, we obtain

$$3zF_z + 5tF_t = 12 \neq 0$$

at w, so $(F_z, F_t) \neq (0,0)$ at w, and W is smooth at w. Thus X is smooth at q. Hence $\operatorname{Sing} X \subset \{p\}$.

Finally, we compute the index of K_X at p. The dualizing sheaf of the hypersurface V is free, generated by the Poincaré residue

$$\eta := \operatorname{Res} \frac{du \wedge dx \wedge dz}{F(u,x,z,1)}$$

(on $V\backslash\{0\}$, which is all that is used, by adjunction [Har77, Proposition II.8.20]). Under $\zeta \in \mu_5$, $du \wedge dx \wedge dz$ is multiplied by ζ^{1+4+3} and $F(u,x,z,1)$ by ζ^{12}, so

$$\zeta^* \eta = \zeta^{1+4+3-12}\eta = \zeta\eta.$$

Let m be a positive integer such that mK_X is Cartier near p, and let ξ be a local generator of $\mathcal{O}_X(mK_X)$ at p. Since $V\backslash\{0\} \to (V\backslash\{0\})/\mu_5$ is étale, the pullback of ξ to a punctured neighborhood of 0 in V is a μ_5-invariant nowhere vanishing section $h\eta^m$ of $\omega_V^{\otimes m}$, where h is a unit on the punctured neighborhood. Since V is a normal surface, the functions h and h^{-1} extend to regular functions near 0 [Har77, Proposition II.6.3A], so $h(0) \neq 0$. Evaluating $\zeta^*(h\eta^m) = h\eta^m$ at 0, we obtain $\zeta^m h(0) = h(0)$. Hence $\zeta^m = 1$ for any $\zeta \in \mu_5$, and $5 \mid m$. Conversely, η^5 is μ_5-invariant and nowhere vanishing, so η^5 descends to a nowhere vanishing section ξ_5 of $\mathcal{O}_X(5K_X)$ on a punctured neighborhood of p. Since X is normal and $\mathcal{O}_X(5K_X)$ is reflexive, ξ_5 extends to a generator of $\mathcal{O}_X(5K_X)$ near p. Thus mK_X is Cartier near p if and only if $5 \mid m$. In particular, K_X is not Cartier at p, so p is a singular point of X, and $\operatorname{Sing} X = \{p\}$. □

The next lemma verifies the conditions on the class group in Lemma 2.4.

Lemma 3.4. *Notation as in Construction 3.1. Then* $\mathrm{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$ *and* $K_X \sim -D$.

Proof. By Lemma 3.2, $X\backslash D \cong \mathbb{A}^2$. Since $\mathbb{C}[a,b]$ is a UFD, $\mathrm{Cl}(X\backslash D) = 0$ [Har77, Proposition II.6.2]. Thus the kernel of the surjection $\mathrm{Cl}(X) \to \mathrm{Cl}(X\backslash D) = 0$, which is $\mathrm{Cl}(X)$, is generated by $[D]$ [Har77, Proposition II.6.5]. Suppose that $nD = \mathrm{div}(\varphi)$ for some integer n and some non-zero rational function φ on X. Then $\varphi|_{X\backslash D}$ is a unit of $\mathbb{C}[a,b]$, so φ is a non-zero constant, and $nD = 0$. Hence $n = 0$, and $\mathrm{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$.

We compute the divisor of the rational two-form $da \wedge db$ on X. Since (a,b) are coordinates of $X\backslash D \cong \mathbb{A}^2$, $da \wedge db$ has neither zeros nor poles on $X\backslash D$, so the divisor of $da \wedge db$ is a multiple of D. By [Ian00, 5.3], $X \cap (x \neq 0)$ is W/μ_4 (Construction 3.1(7)), since $F(u,1,z,t)$ is μ_4-invariant. Since $z^4 = x^3$ on D, we have $D \cap (x \neq 0) = D\backslash\{p\}$, and the preimage of D in W is $W \cap (u = 0)$, on which $z^4 = 1$. Since $\gcd(4,3) = 1$, μ_4 acts freely on $(z \neq 0)$, so the quotient morphism $W \to X \cap (x \neq 0)$ is étale along $W \cap (u = 0)$. On W, we write

$$F_z = 4z(z^2 - ut) - u^9$$

for the partial derivative $\partial F/\partial z$ evaluated at $x = 1$. Along $u = 0$, we have $F_z = 4z^3 \neq 0$. Thus (u,t) are local coordinates of W near any point of $W \cap (u = 0)$, u is a reduced local equation of $W \cap (u = 0)$, and $du \wedge dt$ is a local generator of the canonical sheaf of W there.

On W, we have $a = u^{-4}$ and $b = (z^2 - ut)u^{-6}$. Differentiating the equation of W and taking the wedge product with du, we obtain

$$F_z(du \wedge dz) = 2u(z^2 - ut)(du \wedge dt).$$

Since $da = -4u^{-5}du$ and $4z(z^2 - ut) - F_z = u^9$, we obtain

$$\begin{aligned} da \wedge db &= -4u^{-11}\big(2z(du \wedge dz) - u(du \wedge dt)\big) \\ &= -4u^{-11} \cdot \frac{u\big(4z(z^2 - ut) - F_z\big)}{F_z}(du \wedge dt) = -\frac{4}{F_z}u^{-1}(du \wedge dt). \end{aligned}$$

Since F_z is a unit near $W \cap (u = 0)$, the order of $da \wedge db$ along the preimage of D is -1. Since $W \to X \cap (x \neq 0)$ is étale along the preimage of D, the order of $da \wedge db$ along D is -1. Thus the divisor of $da \wedge db$ is $-D$, and $K_X \sim -D$. □

We obtain the rationality of the singularity p, which is a hypothesis of Question 1.1 and of Lemma 2.4, from global properties of X, without computing a resolution.

Lemma 3.5. *Notation as in Construction 3.1. Then X is a rational surface, and $X \ni p$ is a rational singularity.*

Proof. Let $\pi : Y \to X$ be a projective resolution that is an isomorphism over $X\backslash\{p\}$ (it exists by [KM98, Theorem 0.2], since $\mathrm{Sing}\, X = \{p\}$ by Lemma 3.3), for example the minimal resolution. By Lemma 3.2, $X\backslash D \cong \mathbb{A}^2$, so X is rational. Thus Y is a smooth projective rational surface, and

$$H^1(Y, \mathcal{O}_Y) = H^2(Y, \mathcal{O}_Y) = 0$$

[Har77, Proposition V.3.4, Theorem V.5.5].

Since X is a normal surface, X is Cohen–Macaulay [Mat89, Theorem 23.8]. By Serre duality [KM98, Theorem 5.71, Proposition 5.75] and Lemma 3.4, $H^2(X, \mathcal{O}_X)$ is dual to

$$H^0(X, \mathcal{O}_X(K_X)) = H^0(X, \mathcal{O}_X(-D)).$$

A non-zero element of $H^0(X, \mathcal{O}_X(-D))$ is a regular function on the normal projective variety X vanishing along D, that is, a non-zero constant vanishing along D, which is impossible. Thus $H^2(X, \mathcal{O}_X) = 0$. Since X is normal, $\pi_*\mathcal{O}_Y = \mathcal{O}_X$ [KM98, Definition 5.8], and hence the five-term exact sequence of the Leray spectral sequence [Wei94, Theorem 5.8.3, 5.8.6] contains the exact sequence

$$H^1(Y, \mathcal{O}_Y) \to H^0(X, R^1\pi_*\mathcal{O}_Y) \to H^2(X, \mathcal{O}_X).$$

Hence $H^0(X, R^1\pi_*\mathcal{O}_Y) = 0$. Since π is an isomorphism over $X\backslash\{p\}$, the coherent sheaf $R^1\pi_*\mathcal{O}_Y$ is supported at p, so $R^1\pi_*\mathcal{O}_Y = 0$. Since π is projective and the fibers of π are of dimension at most one, $R^i\pi_*\mathcal{O}_Y = 0$ for any $i \geq 2$ [Har77, Corollary III.11.2]. Thus π is a rational resolution, and $X \ni p$ is a rational singularity [KM98, Definition 5.8, Theorem 5.10]. □

Proposition 3.6. *Notation as in Construction 3.1. Let $\Gamma_V \subset V$ be the curve defined by $z = 0$ and $u^2 = x^3$, and let $\Gamma \subset X$ be the image of Γ_V under the quotient morphism $V \to V/\mu_5 \cong X\backslash(t = 0)$. Then (Γ, p) is a smooth curve germ of X at p. More precisely, the local analytic ring of (Γ, p) is isomorphic to $\mathbb{C}\{s^5\}$, where s is the parameter of the normalization of $(\Gamma_V, 0)$ with $u = s^3$ and $x = s^2$.*

Proof. Let $\mathbb{C}[V] = \mathbb{C}[u, x, z]/(F(u, x, z, 1))$ be the coordinate ring of V, with the action of μ_5 of Construction 3.1(6). Since z is semi-invariant under μ_5, the ideal $(z) \subset \mathbb{C}[V]$ is μ_5-stable. Since μ_5 is finite, the Reynolds operator $\frac{1}{5}\sum_{\zeta\in\mu_5} \zeta^*$ is a $\mathbb{C}[V]^{\mu_5}$-linear projection onto the invariants, so taking μ_5-invariants is exact. Thus

$$\mathbb{C}[V]^{\mu_5}/\big((z) \cap \mathbb{C}[V]^{\mu_5}\big) = \big(\mathbb{C}[V]/(z)\big)^{\mu_5}.$$

The left side is the coordinate ring of the closed subscheme of $V/\mu_5 \cong X\backslash(t = 0)$ defined by the ideal $(z) \cap \mathbb{C}[V]^{\mu_5}$. Since $\mathbb{C}[V]/(z)$ is integral over $(\mathbb{C}[V]/(z))^{\mu_5}$, the underlying set of the closed subscheme defined by $(z) \cap \mathbb{C}[V]^{\mu_5}$ is the image of $V \cap (z = 0)$.

Since $F(u, x, 0, 1) = u^2 - x^3$, we have

$$\mathbb{C}[V]/(z) = \mathbb{C}[u, x]/(u^2 - x^3) \cong \mathbb{C}[s^2, s^3], \qquad u = s^3,\ x = s^2.$$

Thus $V \cap (z = 0) = \Gamma_V$, and s is the parameter of the normalization of Γ_V. The ring $(\mathbb{C}[V]/(z))^{\mu_5}$ is a subring of the reduced ring $\mathbb{C}[V]/(z)$, so $\mathbb{C}[V]^{\mu_5}/((z) \cap \mathbb{C}[V]^{\mu_5})$ is the coordinate ring of Γ with the reduced structure. Since $(\zeta^2)^3 = \zeta$ and $(\zeta^2)^2 = \zeta^4$, $\zeta \in \mu_5$ acts on $\mathbb{C}[s^2, s^3]$ by $s \to \zeta^2 s$. The ring $\mathbb{C}[s^2, s^3]$ is spanned by the monomials s^k with k in the semigroup $\langle 2, 3\rangle$, and $\zeta^* s^k = \zeta^{2k} s^k$. Thus s^k is μ_5-invariant if and only if $5 \mid k$. Since $5 = 2 + 3$ and $\langle 2, 3\rangle$ contains all integers $k \geq 2$, any non-negative multiple of 5 is in $\langle 2, 3\rangle$. Hence

$$\big(\mathbb{C}[V]/(z)\big)^{\mu_5} = \mathbb{C}[s^5], \qquad s^5 = s^3 \cdot s^2 = ux.$$

Therefore, $\Gamma \cong \mathbb{A}^1$ with coordinate ux, and p, the image of 0, corresponds to $s = 0$. In particular, Γ is smooth at p. Thus (Γ, p) is a reduced closed analytic subspace germ of (X, p) of dimension one, and the local analytic ring of (Γ, p) is $\mathbb{C}\{s^5\}$, the ring of convergent power series in the variable $s^5 = ux$, which is isomorphic to $\mathbb{C}\{s\}$. Hence (Γ, p) is a smooth curve germ of X at p (Definition 2.1). □

Proof of Theorem 1.2. The surface X, the point p, and the prime divisor D of Theorem 1.2 are those of Construction 3.1.

(1) By Lemma 3.2, X is a normal projective surface. By Lemma 3.3, p is the unique singular point of X. By Lemma 3.5, X is rational and $X \ni p$ is a rational singularity.

(2) By Lemma 3.2, D is a prime divisor. By Lemma 3.4, $\mathrm{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$ and $-K_X \sim D$. By Lemmas 3.2, 3.3, and 3.5, X is a normal projective surface with rational singularities. Thus Lemma 2.4 implies that D is $\mathbb{Q}$-Cartier and ample, and that K_X is not pseudo-effective.

(3) By Proposition 3.6, p admits a smooth curve germ. By Lemma 3.3, any closed point $q \neq p$ of X is a smooth point of X, and hence admits a smooth curve germ.

(4) By (1) and (2), X and D satisfy the conditions of Lemma 2.4. Thus X contains no closed curve isomorphic to $\mathbb{P}^1$. □

Remark 3.7. Theorem 1.2 answers Question 1.1 negatively also for the stronger reading of the assumption on smooth curve germs that asks, for any closed point q of X, for a closed curve on X passing through q and smooth at q. For $q = p$, the curve Γ of Proposition 3.6 is the image of the closed curve Γ_V under the finite morphism $V \to V/\mu_5$, so Γ is closed in $X\backslash(t = 0)$, and the germ at p of the closure of Γ in X is the smooth curve germ (Γ, p). Thus the closure of Γ in X is

a closed curve on X passing through p and smooth at p. For $q \neq p$, the point q is a smooth point of X by Lemma 3.3. Take a projective embedding of X and a hyperplane H passing through q and not containing the tangent plane of X at q. Then $H \cap X$ is a curve smooth at q, and the irreducible component of $H \cap X$ passing through q is a closed curve on X smooth at q.

4. A second example

In this section, we construct a second surface and prove that it has the properties (1)–(4) of Theorem 1.2 (Proposition 4.4). In contrast with Section 3, the singular point here is described through a resolution: we blow up $\mathbb{P}^2$ ten times and contract ten curves, and the existence of a smooth curve germ at the singular point follows from Lemma 2.3, since one component of the exceptional curve has coefficient one in the fundamental cycle, rather than from an explicit germ.

Construction 4.1. Let $L \subset \mathbb{P}^2$ be a line, and let $Y \to \mathbb{P}^2$ be the composition of the following ten blow-ups of points. At each step, we keep the letters L and E_j for the strict transforms of L and of the earlier exceptional curves.

(1) Blow up a point of L, and let E_1 be the exceptional curve.
(2) Blow up $L \cap E_1$, and let E_2 be the exceptional curve.
(3) Blow up $L \cap E_2$, and let E_3 be the exceptional curve.
(4) For $i = 4, \dots, 8$, blow up a point of E_{i-1} not contained in $L \cup E_1 \cup \cdots \cup E_{i-2}$, and let E_i be the exceptional curve.
(5) Blow up $E_7 \cap E_8$, and let E_9 be the exceptional curve.
(6) Blow up a point of E_9 not contained in $L \cup E_1 \cup \cdots \cup E_8$, and let E_{10} be the exceptional curve.

The points in (1), (4), and (6) may be any points with the indicated properties. Let T and G be the divisors

$$T := L + E_1 + \cdots + E_9,$$
$$G := 3L + 2E_1 + 4E_2 + 6E_3 + 5E_4 + 4E_5 + 3E_6 + 2E_7 + E_8 + 2E_9.$$

After the first eight blow-ups, the curves $L, E_1, \dots, E_7$ form a configuration of (-2)-curves of type E_8, and the last two blow-ups move E_8, the only component of G with coefficient one, into T, so that E_{10} is the only component of $E_{10} + G \sim -K_Y$ not contained in $\operatorname{Supp} T$ (Lemma 4.2(3)).

Lemma 4.2. *Notation as in Construction 4.1. Then the following hold.*

(1) The curves $L, E_1, \dots, E_{10}$ are smooth rational curves, $(E_7^2) = -3$, $(E_{10}^2) = -1$, and $(C^2) = -2$ for any irreducible component $C \neq E_7$ of T.
(2) Two distinct curves C, C' among $L, E_1, \dots, E_{10}$ intersect if and only if $\{C, C'\}$ is one of the ten pairs $\{L, E_3\}$, $\{E_1, E_2\}$, $\{E_2, E_3\}$, $\{E_3, E_4\}$, $\{E_4, E_5\}$, $\{E_5, E_6\}$, $\{E_6, E_7\}$, $\{E_7, E_9\}$, $\{E_8, E_9\}$, $\{E_9, E_{10}\}$. If C and C' intersect, then they intersect transversally at one point.
(3) $-K_Y \sim E_{10} + G$, $(G \cdot E_7) = (G \cdot E_9) = -1$, and $(G \cdot C) = 0$ for any other irreducible component C of T.
(4) The intersection matrix of the irreducible components of T is negative definite.
(5) The classes of $L, E_1, \dots, E_{10}$ form a basis of the free abelian group $\operatorname{Pic}(Y)$.

Proof. For $i = 1, \dots, 10$, let $f_i : Y_i \to Y_{i-1}$ be the i-th blow-up in Construction 4.1, where $Y_0 = \mathbb{P}^2$ and $Y_{10} = Y$, and let $q_i \in Y_{i-1}$ be the center of f_i. For a smooth curve $C \subset Y_{i-1}$, let $\tilde{C} \subset Y_i$ be the strict transform of C, and let $m_C := 1$ if $q_i \in C$ and $m_C := 0$ otherwise. If $q_i \notin C$, then f_i is an isomorphism over a neighborhood of C, so $\tilde{C} \to C$ is an isomorphism. If $q_i \in C$, then $\tilde{C}$ is the blowing-up of C along the ideal sheaf of q_i in $\mathcal{O}_C$ [Har77, Corollary II.7.15], which is invertible since C is a smooth curve, so $\tilde{C} \to C$ is an isomorphism [Har77, Proposition II.7.14].

Moreover, $E_i \cong \mathbb{P}^1$, and for distinct smooth curves $C, C' \subset Y_{i-1}$, we have

$$f_i^* C = \tilde{C} + m_C E_i, \quad (E_i^2) = -1, \quad (\tilde{C}^2) = (C^2) - m_C,$$
$$(\tilde{C} \cdot E_i) = m_C, \quad (\tilde{C} \cdot \tilde{C}') = (C \cdot C') - m_C m_{C'}$$

[Har77, Propositions V.3.1, V.3.2, V.3.6, Corollary V.3.7]. In the notation of Construction 4.1, $\tilde{C}$ keeps the letter of C. Moreover, two distinct smooth curves C, C' on a smooth projective surface with $(C \cdot C') = 0$ are disjoint, and with $(C \cdot C') = 1$ they intersect transversally at one point [Har77, Proposition V.1.4].

We prove (1) and (2). Since $L \cong \mathbb{P}^1$, the curves $L, E_1, \dots, E_{10}$ are smooth rational curves on any Y_i where they are defined. We compute the intersection numbers among $L, E_1, \dots, E_i$ on Y_i by induction on i, starting from $(L^2) = 1$ on $\mathbb{P}^2$. On Y_1, we have $(L^2) = 0$, $(E_1^2) = -1$, and $(L \cdot E_1) = 1$. Thus $L \cap E_1 = \{q_2\}$, and on Y_2,

$$(L^2) = -1, \quad (E_1^2) = -2, \quad (E_2^2) = -1, \quad (L \cdot E_1) = 0, \quad (L \cdot E_2) = (E_1 \cdot E_2) = 1.$$

Thus $L \cap E_2 = \{q_3\}$ and $q_3 \notin E_1$, and on Y_3,

$$(L^2) = (E_1^2) = (E_2^2) = -2, \quad (E_3^2) = -1, \quad (L \cdot E_3) = (E_2 \cdot E_3) = (E_1 \cdot E_2) = 1,$$

and $(L \cdot E_1) = (L \cdot E_2) = (E_1 \cdot E_3) = 0$. For $i = 4, \dots, 8$, the point q_i is contained in E_{i-1} and in no other curve among $L, E_1, \dots, E_{i-2}$. Thus on Y_i, $(E_{i-1}^2) = -2$, $(E_i^2) = -1$, $(E_{i-1} \cdot E_i) = 1$, $(C \cdot E_i) = 0$ for any curve C among $L, E_1, \dots, E_{i-2}$, and the other intersection numbers do not change. Hence on Y_8, the curves $L, E_1, \dots, E_7$ have self-intersection -2, $(E_8^2) = -1$, and the pairs of distinct curves with intersection number 1 are

$$\{L, E_3\}, \ \{E_1, E_2\}, \ \{E_2, E_3\}, \ \{E_3, E_4\}, \ \{E_4, E_5\}, \ \{E_5, E_6\}, \ \{E_6, E_7\}, \ \{E_7, E_8\},$$

while any other pair has intersection number 0. In particular, $E_7 \cap E_8 = \{q_9\}$, and q_9 is not contained in any other curve among $L, E_1, \dots, E_6$, because E_8 is disjoint from them. Thus on Y_9,

$$(E_7^2) = -3, \quad (E_8^2) = -2, \quad (E_9^2) = -1, \quad (E_7 \cdot E_8) = 0, \quad (E_7 \cdot E_9) = (E_8 \cdot E_9) = 1,$$

and E_9 is disjoint from $L, E_1, \dots, E_6$. Finally, q_{10} is contained in E_9 and in no other curve among $L, E_1, \dots, E_8$, so on $Y_{10} = Y$, $(E_9^2) = -2$, $(E_{10}^2) = -1$, $(E_9 \cdot E_{10}) = 1$, and E_{10} is disjoint from $L, E_1, \dots, E_8$. This proves the self-intersection numbers in (1), and shows that on Y, two distinct curves C, C' among $L, E_1, \dots, E_{10}$ satisfy $(C \cdot C') = 1$ if $\{C, C'\}$ is one of the ten pairs in (2), and $(C \cdot C') = 0$ otherwise. Thus (2) follows from [Har77, Proposition V.1.4].

We prove (3). For $i = 1, \dots, 10$, we have $K_{Y_i} = f_i^* K_{Y_{i-1}} + E_i$ [Har77, Proposition V.3.3]. If $-K_{Y_{i-1}} \sim \sum_C a_C C$, where C runs over $L, E_1, \dots, E_{i-1}$, then

$$-K_{Y_i} \sim \sum_C a_C \tilde{C} + \Big(\sum_C a_C m_C - 1 \Big) E_i.$$

Thus a blow-up at a point contained in exactly one curve C creates the coefficient $a_C - 1$ on E_i, and a blow-up at the intersection point of two curves C, C' creates the coefficient $a_C + a_{C'} - 1$. Starting from $-K_{\mathbb{P}^2} \sim 3L$, the coefficients of L stay 3, and by the positions of the centers q_i determined in the proof of (1) and (2), the coefficients of $E_1, \dots, E_{10}$ in chronological order are

$$3 - 1 = 2, \ 3 + 2 - 1 = 4, \ 3 + 4 - 1 = 6, \ 6 - 1 = 5, \ 5 - 1 = 4,$$
$$4 - 1 = 3, \ 3 - 1 = 2, \ 2 - 1 = 1, \ 2 + 1 - 1 = 2, \ 2 - 1 = 1.$$

Thus $-K_Y \sim E_{10} + G$. For any irreducible component C of T, $(G \cdot C)$ is (C^2) times the coefficient of C in G, plus the sum of the coefficients in G of the components of T intersecting C. By (1)

and (2), we obtain

$$(G\cdot L)=-6+6=0, \quad (G\cdot E_1)=-4+4=0, \quad (G\cdot E_2)=-8+2+6=0,$$
$$(G\cdot E_3)=-12+3+4+5=0, \quad (G\cdot E_4)=-10+6+4=0, \quad (G\cdot E_5)=-8+5+3=0,$$
$$(G\cdot E_6)=-6+4+2=0, \quad (G\cdot E_7)=-6+3+2=-1, \quad (G\cdot E_8)=-2+2=0,$$
$$(G\cdot E_9)=-4+2+1=-1.$$

This proves (3).

We prove (4). By (1) and (2), removing the row and the column of E_3 from minus the intersection matrix of the irreducible components of T leaves a block diagonal matrix with three blocks, corresponding to L, to the ordered pair (E_2, E_1), and to the ordered tuple $(E_4, E_5, E_6, E_7, E_9, E_8)$. Each block is tridiagonal with off-diagonal entries -1, and the diagonal entries are (2), $(2,2)$, and $(2,2,2,3,2,2)$ respectively. The leading principal minors of the three blocks are

$$2; \qquad 2,\ 3; \qquad 2,\ 3,\ 4,\ 9,\ 14,\ 19,$$

so the three blocks are positive definite by Sylvester's criterion. The curve E_3 intersects exactly L, E_2, and E_4 among the other components of T, which are the first vertices of the three blocks. The diagonal entries at L, E_2, and E_4 of the inverses of the three blocks are $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{15}{19}$, where 15 is the determinant of the tridiagonal matrix with diagonal $(2,2,3,2,2)$ (the leading principal minors of the matrix are $2,3,7,11,15$). Thus the Schur complement at E_3 is

$$2-\frac{1}{2}-\frac{2}{3}-\frac{15}{19}=\frac{5}{114}>0.$$

Therefore minus the intersection matrix is positive definite.

We prove (5). The class $[L]$ is a basis of $\operatorname{Pic}(\mathbb{P}^2)\cong\mathbb{Z}$. For $i=1,\dots,10$, we have

$$\operatorname{Pic}(Y_i)=f_i^*\operatorname{Pic}(Y_{i-1})\oplus\mathbb{Z}[E_i]$$

[Har77, Proposition V.3.2]. If the classes of $L, E_1, \dots, E_{i-1}$ form a basis of $\operatorname{Pic}(Y_{i-1})$, then the classes of $f_i^*L, f_i^*E_1, \dots, f_i^*E_{i-1}, E_i$ form a basis of $\operatorname{Pic}(Y_i)$, and $[\tilde{C}]=f_i^*[C]-m_C[E_i]$ for any curve C among $L, E_1, \dots, E_{i-1}$. Thus the change from the basis $f_i^*[L], f_i^*[E_1], \dots, f_i^*[E_{i-1}], [E_i]$ to the classes of $\tilde{L}, \tilde{E}_1, \dots, \tilde{E}_{i-1}, E_i$ is given by a triangular integral matrix with determinant one, and (5) follows by induction on i. □

Since $(E_7^2)=-3$, T does not contract to a Du Val singularity (a contraction of T would be the minimal resolution, since T contains no (-1)-curve, and $(K_Y\cdot E_7)=1$ by adjunction, while a Du Val singularity has $(K\cdot E)=0$ for any exceptional curve E of the minimal resolution [KM98, Definition 4.4]), and we verify the conditions of [Lip69, Theorem 27.1] through the bound of the fundamental cycle of T by the divisor G of Lemma 4.2(3), which also shows that the coefficient of E_8 in the fundamental cycle of T is one.

Construction-Proposition 4.3. Notation as in Construction 4.1. Then there exist a birational morphism $\pi: Y\to X$ onto a normal projective surface X and a closed point $p\in X$ such that $\pi^{-1}(p)=\operatorname{Supp}T$ and π is an isomorphism over $X\backslash\{p\}$. Moreover, p is the unique singular point of X, $X\ni p$ is a rational singularity, and the fundamental cycle F_p of $X\ni p$ on Y satisfies $F_p\le G$ and $\operatorname{mult}_{E_8}F_p=1$.

Proof. By Lemma 4.2(2), $\operatorname{Supp}T$ is connected, and by Lemma 4.2(4), the intersection matrix of the irreducible components of T is negative definite. Let F be the fundamental cycle of T (Definition 2.2). Then the coefficient of any irreducible component of T in F is at least 1 (Definition 2.2). The divisor G is non-zero with non-negative integer coefficients, and $(G\cdot C)\le 0$ for any irreducible component C of T (Lemma 4.2(3)). By the minimality of F, we have

$$0<F\le G.$$

The divisor F is the fundamental curve for $\operatorname{Supp}T$ in [Lip69, p. 275 and Theorem 27.1], since Lipman defines the fundamental curve as the smallest non-zero divisor $F'\ge 0$ supported on

$\operatorname{Supp} T$ such that $(F' \cdot C) \leq 0$ for any irreducible component C of T, which is the defining property of F in Definition 2.2.

We show that $\chi(\mathcal{O}_F) > 0$. Since Y is a smooth projective rational surface, $H^1(Y, \mathcal{O}_Y) = 0$ [Har77, Proposition V.3.4, Theorem V.5.5]. Since $K_Y + G \sim -E_{10}$ (Lemma 4.2(3)) and $E_{10} \geq 0$ is a non-zero divisor on the projective surface Y, Serre duality [Har77, Corollary III.7.7] implies

$$H^2(Y, \mathcal{O}_Y(-G)) \cong H^0(Y, \mathcal{O}_Y(K_Y + G))^\vee \cong H^0(Y, \mathcal{O}_Y(-E_{10}))^\vee = 0.$$

Thus the exact sequence

$$0 \to \mathcal{O}_Y(-G) \to \mathcal{O}_Y \to \mathcal{O}_G \to 0$$

implies $H^1(\mathcal{O}_G) = 0$. Since $F \leq G$, there exists a surjection $\mathcal{O}_G \to \mathcal{O}_F$. The kernel of $\mathcal{O}_G \to \mathcal{O}_F$ is a sheaf on G, and G is a noetherian topological space of dimension one, so the second cohomology group of the kernel vanishes [Har77, Theorem III.2.7]. Hence $H^1(\mathcal{O}_G) \to H^1(\mathcal{O}_F)$ is surjective, and $H^1(\mathcal{O}_F) = 0$. Since $F \neq 0$, we have $h^0(\mathcal{O}_F) \geq 1$. Therefore

$$\chi(\mathcal{O}_F) = h^0(\mathcal{O}_F) \geq 1 > 0.$$

The structure morphism $Y \to \operatorname{Spec} \mathbb{C}$ is projective, and the irreducible components of T are distinct integral curves with exceptional support relative to $Y \to \operatorname{Spec} \mathbb{C}$ in the sense of [Lip69, Section 12], since they are proper over $\operatorname{Spec} \mathbb{C}$, which is a point. Moreover, Y is normal at every point of $\operatorname{Supp} T$ since Y is smooth. The normality of Y at the points of $\operatorname{Supp} T$, the connectedness of $\operatorname{Supp} T$, and the negative definiteness of the intersection matrix of the irreducible components of T shown at the beginning of the proof are the remaining hypotheses of [Lip69, Theorem 27.1], and the condition there for the existence of the contraction is $\chi(\mathcal{O}_F) > 0$. Since $\chi(\mathcal{O}_F) > 0$, by [Lip69, Theorem 27.1], there exist a projective scheme X and a proper morphism $\pi : Y \to X$ which contracts $\operatorname{Supp} T$ rationally to a point p. By the definitions on [Lip69, p. 275], this means the following: $\pi(\operatorname{Supp} T) = \{p\}$, p is a normal point of X, π induces an isomorphism $Y \setminus \operatorname{Supp} T \to X \setminus \{p\}$, and $R^1\pi_*\mathcal{O}_Y = 0$.

Thus π is surjective, $\pi^{-1}(p) = \operatorname{Supp} T$, and π is an isomorphism over $X \setminus \{p\}$. Since $X \setminus \{p\}$ is isomorphic to the smooth surface $Y \setminus \operatorname{Supp} T$ and p is a normal point, X is a normal projective surface, and π is birational. Since X is normal, $\pi_*\mathcal{O}_Y = \mathcal{O}_X$ [KM98, Definition 5.8]. Since Y is projective, π is projective, and the fibers of π are of dimension at most one, so $R^i\pi_*\mathcal{O}_Y = 0$ for any $i \geq 2$ [Har77, Corollary III.11.2]. Since moreover $R^1\pi_*\mathcal{O}_Y = 0$, π is a rational resolution, and $X \ni p$ is a rational singularity [KM98, Definition 5.8, Theorem 5.10].

We show that p is a singular point of X. Suppose that p is a smooth point of X. Then X is smooth, and π is a birational morphism between smooth projective surfaces which is not an isomorphism, since $\pi^{-1}(p) = \operatorname{Supp} T$ is a curve. By [Har77, Corollary V.5.4], π is a composition of blow-ups of points, so the exceptional curve of the last blow-up is a curve $C \subset Y$ with $(C^2) = -1$ and $\pi(C)$ is a point. Then $C \subset \pi^{-1}(p) = \operatorname{Supp} T$, which contradicts Lemma 4.2(1), since any irreducible component C of T satisfies $(C^2) \leq -2$. Thus p is a singular point of X, and p is the unique singular point of X.

Since π is a resolution of singularities of X which is an isomorphism over the smooth locus $X \setminus \{p\}$ and $\pi^{-1}(p) = \operatorname{Supp} T$, we have $F_p = F$ (Definition 2.2). Thus $F_p \leq G$, and since the coefficient of E_8 in G is 1,

$$1 \leq \operatorname{mult}_{E_8} F_p \leq \operatorname{mult}_{E_8} G = 1,$$

where the first inequality follows from Definition 2.2. Hence $\operatorname{mult}_{E_8} F_p = 1$. □

Proposition 4.4. *Notation as in Construction-Proposition 4.3, and let $D := \pi(E_{10})$ with the reduced structure. Then the following hold.*

1. *X is a normal projective rational surface, p is the unique singular point of X, and $X \ni p$ is a rational singularity.*
2. *D is a prime divisor, D is $\mathbb{Q}$-Cartier and ample, $\operatorname{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$, and $-K_X \sim D$. In particular, K_X is not pseudo-effective.*
3. *Any closed point of X admits a smooth curve germ.*

(4) X contains no closed curve isomorphic to $\mathbb{P}^1$.

Proof. (1) By Construction-Proposition 4.3, X is a normal projective surface, p is the unique singular point of X, and $X \ni p$ is a rational singularity. Since π is birational and $Y \to \mathbb{P}^2$ is a composition of blow-ups of points (Construction 4.1), X is rational.

(2) Since E_{10} is an integral curve distinct from the irreducible components $L, E_1, \dots, E_9$ of T (Construction 4.1), $E_{10} \not\subset \operatorname{Supp} T$, and since π is an isomorphism over $X\backslash\{p\}$, D is a prime divisor, and π restricts to an isomorphism $E_{10}\backslash \operatorname{Supp} T \to D\backslash\{p\}$. Since $\{p\}$ has codimension two in X and $\operatorname{Supp} T$ is the union of the irreducible components C of T, we have

$$\operatorname{Cl}(X) \cong \operatorname{Cl}(X\backslash\{p\}) \cong \operatorname{Cl}(Y\backslash \operatorname{Supp} T) \cong \operatorname{Pic}(Y) \Big/ \sum_C \mathbb{Z}[C]$$

[Har77, Proposition II.6.5, Corollary II.6.16], where the first and the last isomorphisms are induced by restriction and the second by π. The composite isomorphism sends $[D]$ to $[E_{10}]$. By Lemma 4.2(5), the right-hand side is the free abelian group generated by $[E_{10}]$. Thus $\operatorname{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$. Moreover, the restriction of K_X to $X\backslash\{p\}$ is a canonical divisor of $X\backslash\{p\}$, and the restriction of $-K_Y \sim E_{10} + G$ (Lemma 4.2(3)) to $Y\backslash \operatorname{Supp} T$ is linearly equivalent to $E_{10}|_{Y\backslash \operatorname{Supp} T}$. Thus $-K_X|_{X\backslash\{p\}} \sim D|_{X\backslash\{p\}}$, and since restriction $\operatorname{Cl}(X) \to \operatorname{Cl}(X\backslash\{p\})$ is an isomorphism, $-K_X \sim D$.

By (1), X is a normal projective surface with rational singularities. Since D is a prime divisor, $-K_X \sim D$, and $\operatorname{Cl}(X) = \mathbb{Z}[D] \cong \mathbb{Z}$, Lemma 2.4 implies that D is $\mathbb{Q}$-Cartier and ample, and that K_X is not pseudo-effective. This proves (2).

(3) Let $x \in X$ be a closed point. If $x \neq p$, then x is a smooth point of X by (1), and hence admits a smooth curve germ. If $x = p$, then π is a resolution of singularities of X which is an isomorphism over the smooth locus $X\backslash\{p\}$, $X \ni p$ is a rational singularity, and $\operatorname{mult}_{E_8} F_p = 1$, where E_8 is an irreducible component of $\pi^{-1}(p) = \operatorname{Supp} T$ (Construction-Proposition 4.3). Thus p admits a smooth curve germ by Lemma 2.3.

(4) By (1) and (2), X and D satisfy the conditions of Lemma 2.4. Thus X contains no closed curve isomorphic to $\mathbb{P}^1$. □

References

[Art66] M. Artin, *On isolated rational singularities of surfaces*, Amer. J. Math. **88** (1966), 129–136.

[GSLJ94] G. Gonzalez-Sprinberg and M. Lejeune-Jalabert, *Courbes lisses sur les singularités de surface*, C. R. Acad. Sci. Paris Sér. I Math. **318** (1994), no. 7, 653–656.

[GSLJ97] G. Gonzalez-Sprinberg and M. Lejeune-Jalabert, *Families of smooth curves on surface singularities and wedges*, Ann. Polon. Math. **67** (1997), no. 2, 179–190.

[Har77] R. Hartshorne, *Algebraic geometry*, Grad. Texts in Math. **52**, Springer-Verlag, New York–Heidelberg (1977).

[Ian00] A. R. Iano-Fletcher, *Working with weighted complete intersections*, in: *Explicit birational geometry of 3-folds*, London Math. Soc. Lecture Note Ser. **281**, Cambridge Univ. Press, Cambridge (2000), 101–173.

[KM99] S. Keel and J. McKernan, *Rational curves on quasi-projective surfaces*, Mem. Amer. Math. Soc. **140** (1999), no. 669, viii+153 pp.

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergeb. Math. Grenzgeb. (3) **48**, Springer-Verlag, Berlin (2004).

[Lip69] J. Lipman, *Rational singularities, with applications to algebraic surfaces and unique factorization*, Publ. Math. Inst. Hautes Études Sci. **36** (1969), 195–279.

[Mat89] H. Matsumura, *Commutative ring theory*, second ed., Cambridge Stud. Adv. Math. **8**, Cambridge Univ. Press, Cambridge (1989).

[PP24] K. Palka and T. Pełka, *Classification of del Pezzo surfaces of rank one. I. Height 1 and 2, II. Descendants with elliptic boundaries*, arXiv:2412.21174.

[Wei94] C. A. Weibel, *An introduction to homological algebra*, Cambridge Stud. Adv. Math. **38**, Cambridge Univ. Press, Cambridge (1994).

[Zhu18] Z. Zhuang, *Smooth rational curves on singular rational surfaces*, Michigan Math. J. **67** (2018), no. 1, 83–98.

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